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11.1.1 Linking the Singular Value Decomposition to the Spectral Decomposition

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Video



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able to compute the 2-norm of a matrix and the associated vector that generates that 2-norm. And there is a relationship between finding that and finding the eigenvalues and eigenvectors of a related matrix. And therefore the fact that finding the eigenvalues and eigenvectors is inherently difficult. And we can come up with algorithms that converge to them, but we can't really write down an

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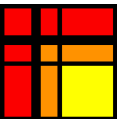
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Reading assignment

1/1 point (graded)



Read Unit 11.1.1 of the notes. [\[LINK\]](#)

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Homework 11.1.1.1

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$ and $A = U \Sigma V^H$ its SVD, where Σ has the structure

$$\Sigma = \left(\begin{array}{c|c} \Sigma_{TL} & 0_{r \times (n-r)} \\ \hline 0_{(m-r) \times r} & 0_{(m-r) \times (n-r)} \end{array} \right), \text{ with } \Sigma_{TL} = \text{diag}(\sigma_0, \dots, \sigma_{r-1} \text{ and } \sigma_0 \geq \sigma_1 \geq \dots \geq \sigma_{r-1}$$

Give the Spectral Decomposition of the matrix $A^H A$.

☒ done



Solution:

For a complete solution, see [the notes](#).

There is a video in the notes.

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 Answers are displayed within the problem

Homework 11.1.1.2

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$ and $A = U \Sigma V^H$ its SVD, where Σ has the structure

$$\Sigma = \left(\begin{array}{c|c} \Sigma_{TL} & 0_{r \times (n-r)} \\ \hline 0_{(m-r) \times r} & 0_{(m-r) \times (n-r)} \end{array} \right), \text{ with } \Sigma_{TL} = \text{diag}(\sigma_0, \dots, \sigma_{r-1} \text{ and } \sigma_0 \geq \sigma_1 \geq \dots \geq \sigma_{r-1}$$

Give the Spectral Decomposition of the matrix AA^H .

☒ done



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